

Home SOC Lab

Security Monitoring & Threat Detection Project

Prepared by: Raghad Alanazi

Platform: Splunk Enterprise 10.4.0 | Environment: Windows 11 (VM)

June 2026

1. Executive Summary

This report documents the design, implementation, and validation of a home Security Operations Center (SOC) lab built to demonstrate practical, hands-on threat detection and monitoring capabilities. Splunk Enterprise was deployed as the SIEM platform and configured to ingest Windows Security event logs from a Windows 11 virtual machine acting as the monitored endpoint.

Three realistic attack scenarios were simulated on the endpoint: a brute-force authentication attempt, suspicious PowerShell execution consistent with post-exploitation reconnaissance, and a port-scanning activity consistent with service discovery. For each scenario, detection logic was developed using Splunk Search Processing Language (SPL), validated against the resulting event data, and converted into scheduled, enabled alerts.

The three detections were consolidated into a SOC monitoring dashboard, and the full sequence of events was reconstructed into an incident investigation timeline mapped to the MITRE ATT&CK framework. Together, this project demonstrates practical capability across log source configuration, detection engineering, SPL query development, alerting, dashboarding, and incident investigation core competencies for a SOC Analyst or Cybersecurity Analyst role.

2. Lab Environment & Tools

The table below summarizes the components and configuration used to build the lab environment.

Component	Description
Host	macOS with VMware Fusion
Monitored Endpoint	Windows 11 (64-bit ARM) virtual machine
SIEM Platform	Splunk Enterprise (Free Trial), version 10.4.0
Log Sources	Windows Event Logs Security, System, Application (collected via local Windows Event Log monitoring input)
Auditing Configuration	Windows Security Auditing enabled for Process Creation (Event ID 4688) and Filtering Platform Connection (Event ID 5158); PowerShell Script Block Logging enabled
Attack Simulation	PowerShell brute-force authentication loop, reconnaissance command execution, and TCP port scan loop against the local host

3. Detection Scenario 1: Brute Force Authentication Attempts

Repeated failed logon attempts were simulated against a non-existent account (“FakeUser”) using the Windows net use command with intentionally incorrect credentials. This mimics a password-guessing / brute-force attack against the host.

3.1 Raw Event Search

```
index=* sourcetype="WinEventLog:Security" EventCode=4625
```

The screenshot shows the Splunk Enterprise interface with a search query: `index=* sourcetype="WinEventLog:Security" EventCode=4625`. The search results show 4 events. The first event is expanded, showing details for a failed logon attempt on 6/13/2026 at 04:21:53.962 AM. The event details include: LogName=Security, EventCode=4625, EventType=0, ComputerName=Taif, and host = Taif. The source is WinEventLog:Security and the sourcetype is WinEventLog:Security. The second event is also visible, showing a failed logon attempt on 6/13/2026 at 04:21:18.034 AM with similar details.

Time	Event
6/13/2026 04:21:53.962 AM	LogName=Security EventCode=4625 EventType=0 ComputerName=Taif host = Taif source = WinEventLog:Security sourcetype = WinEventLog:Security
6/13/2026 04:21:18.034 AM	LogName=Security EventCode=4625 EventType=0 ComputerName=Taif

Figure 1: Failed logon events (Event ID 4625) recorded for the FakeUser account

3.2 Detection Query

The detection logic counts failed logon attempts per account and flags any account exceeding three failures:

```
index=* sourcetype="WinEventLog:Security" EventCode=4625 | stats count by Account_Name | where count > 3
```

Result: 4 failed logon events were recorded against the FakeUser account, exceeding the threshold of 3 and triggering Alert 1 “Brute Force Attack - Failed Logins”.

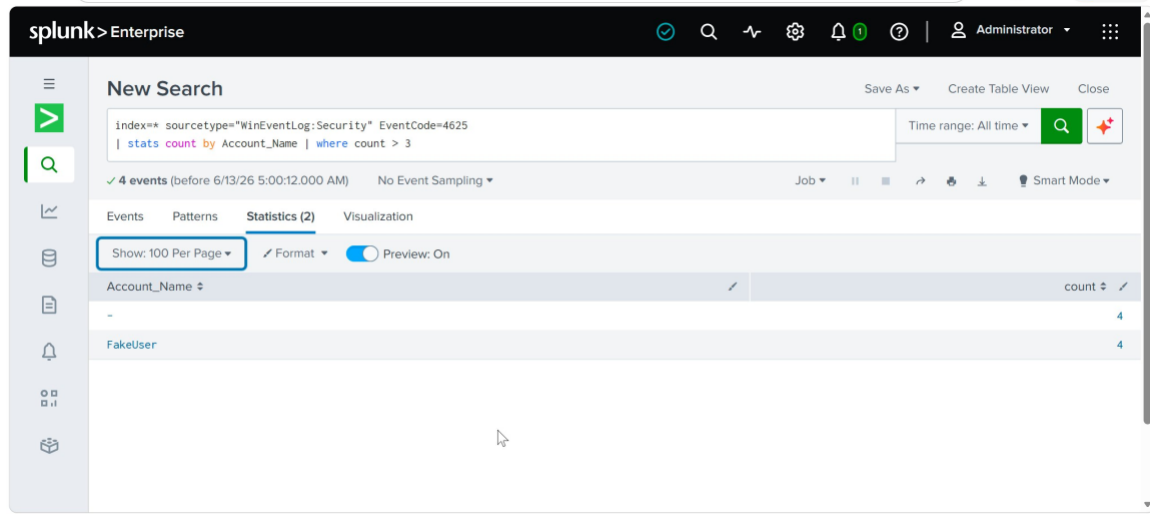


Figure 2: Detection query results - FakeUser account with 4 failed logon attempts

4. Detection Scenario 2: Suspicious PowerShell Execution

PowerShell was executed with the execution policy bypassed (-ExecutionPolicy Bypass) to run host reconnaissance commands (whoami, hostname, ipconfig). This pattern bypassing execution policy to run enumeration commands is commonly associated with post-exploitation activity.

4.1 Raw Event Search

All process creation events (Event ID 4688) were reviewed first to confirm log coverage:

```
index=* sourcetype="WinEventLog:Security" EventCode=4688
```

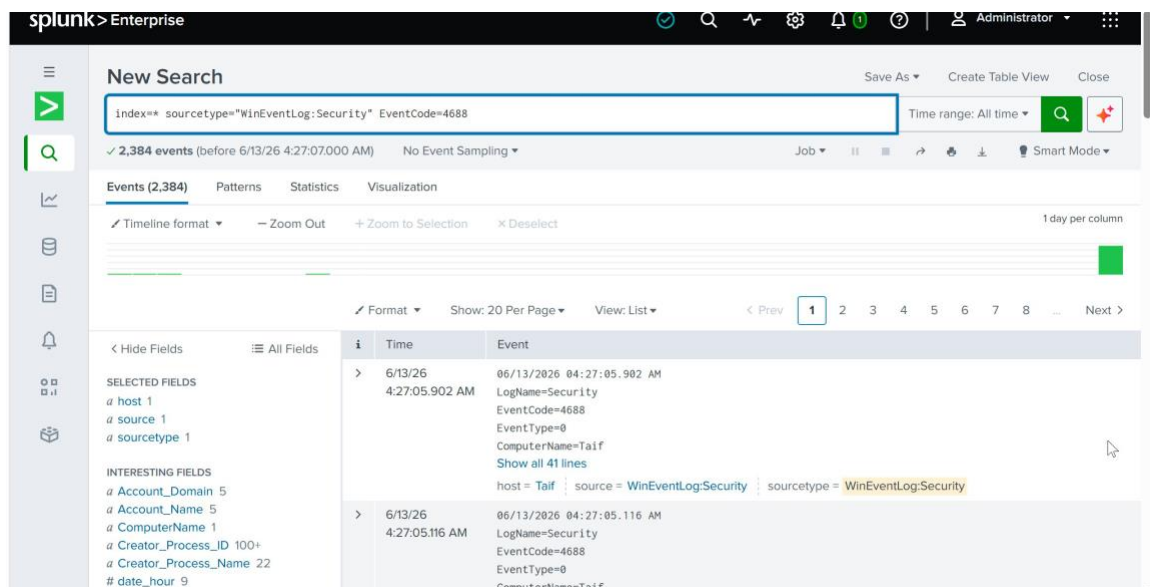


Figure 3: Process creation events (Event ID 4688) - 2,384 total events captured

The search was then narrowed to PowerShell-specific process creations:

```
index=* sourcetype="WinEventLog:Security" EventCode=4688 New_Process_Name="*powershell*"
```

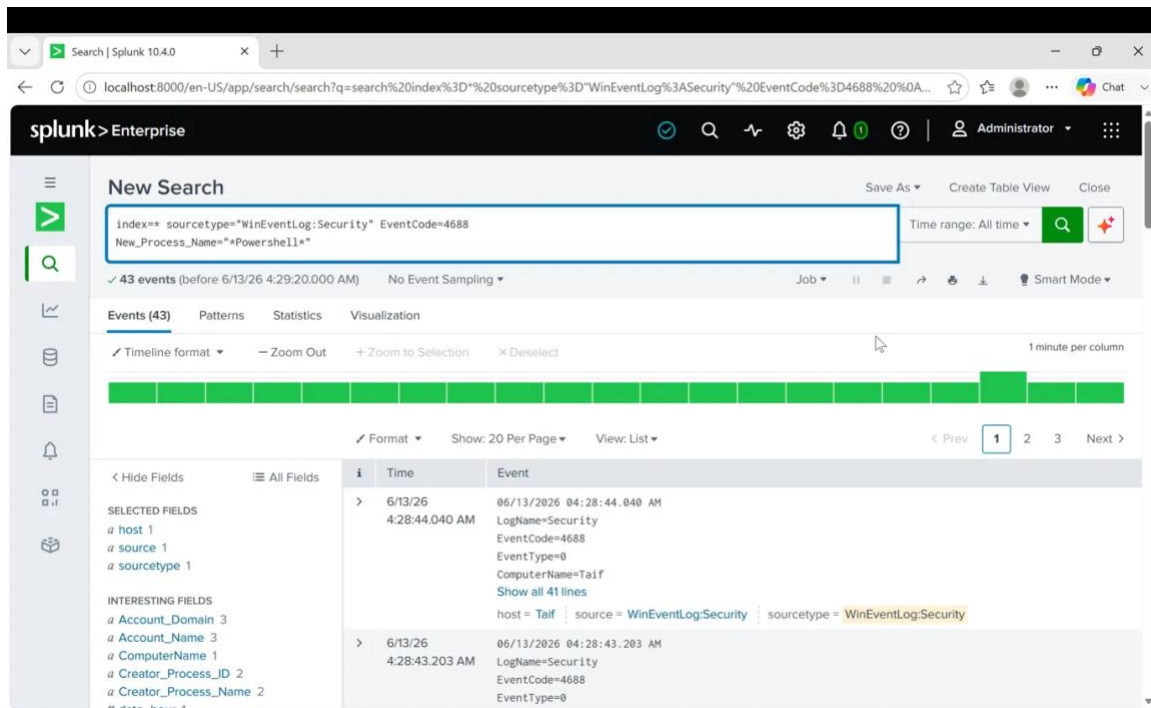


Figure 4: PowerShell process creation events - 43 events identified

4.2 Detection Query

The detection logic groups PowerShell executions by account to identify which user context launched PowerShell:

```
index=* sourcetype="WinEventLog:Security" EventCode=4688 New_Process_Name="*powershell*" | stats count by Account_Name | where count > 0
```

Result: of 120 grouped events, 2 PowerShell executions were attributed to the interactive user account (taifs), with the remainder attributed to the Splunk service account (Splunkd) performing its own monitoring activity. This confirms Alert 2 “Suspicious PowerShell Execution” correctly isolates the user-initiated activity.

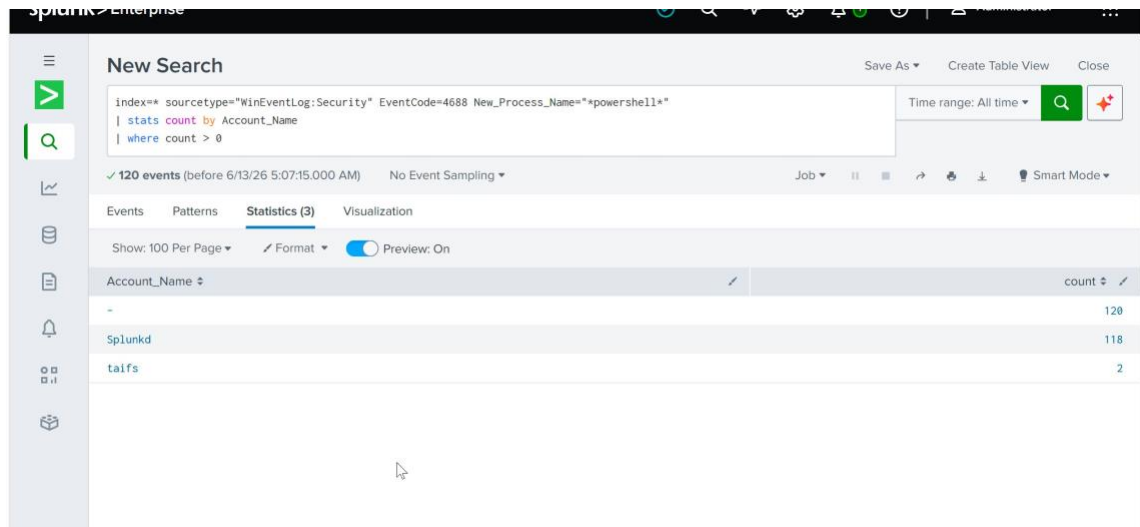


Figure 5: PowerShell executions grouped by account - taifs: 2, Splunkd: 118

5. Detection Scenario 3: Port Scan / Service Discovery

A PowerShell loop attempted TCP connections to ports 1-1024 on the local host. This simulates a port scan, a technique attackers use for service discovery and to plan lateral movement once inside a network.

5.1 Raw Event Search

index=* sourcetype="WinEventLog:Security" EventCode=5158

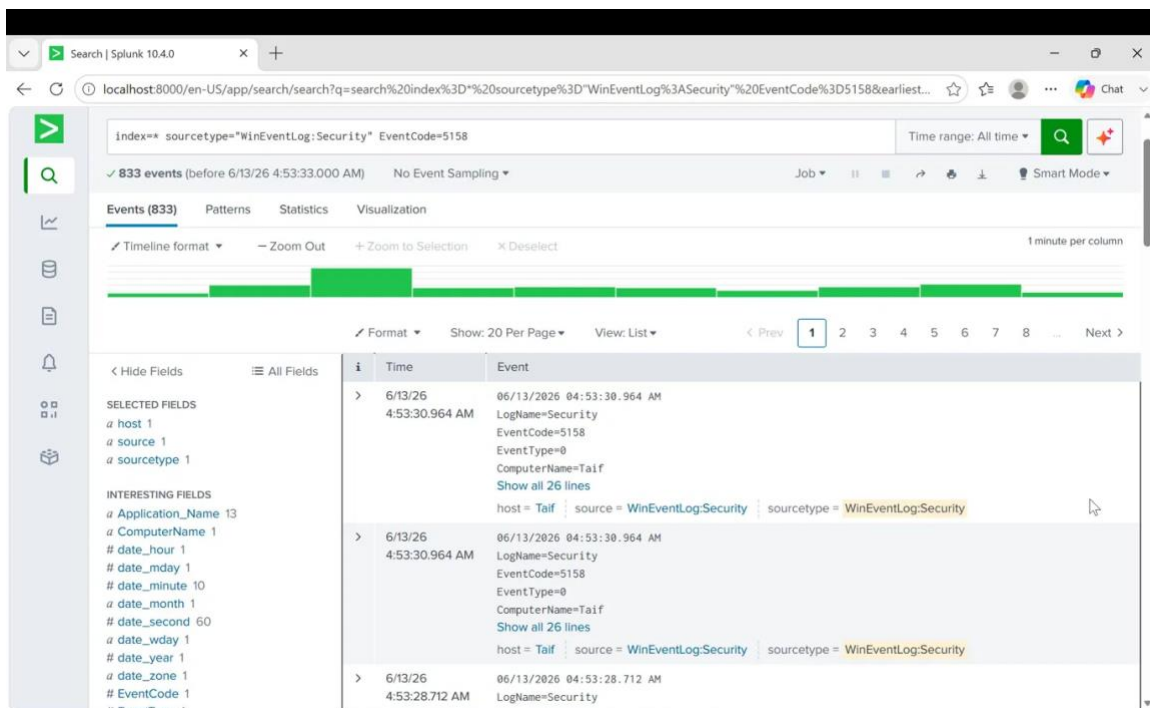


Figure 6: Network connection permit events (Event ID 5158) - 833 total events captured

5.2 Detection Query

The detection logic identifies connection-binding activity originating from shell processes (PowerShell or cmd), which is unusual outside of administrative or scripted tasks:

```
index=* sourcetype="WinEventLog:Security" EventCode=5158 Application_Name="*powershell*" OR  
Application_Name="*cmd*" | stats count by Application_Name
```

Result: 58 connection-binding events were attributed to powershell.exe within a one-hour window, consistent with the simulated port scan and triggering Alert 3 “Suspicious Port Scan - PowerShell”.

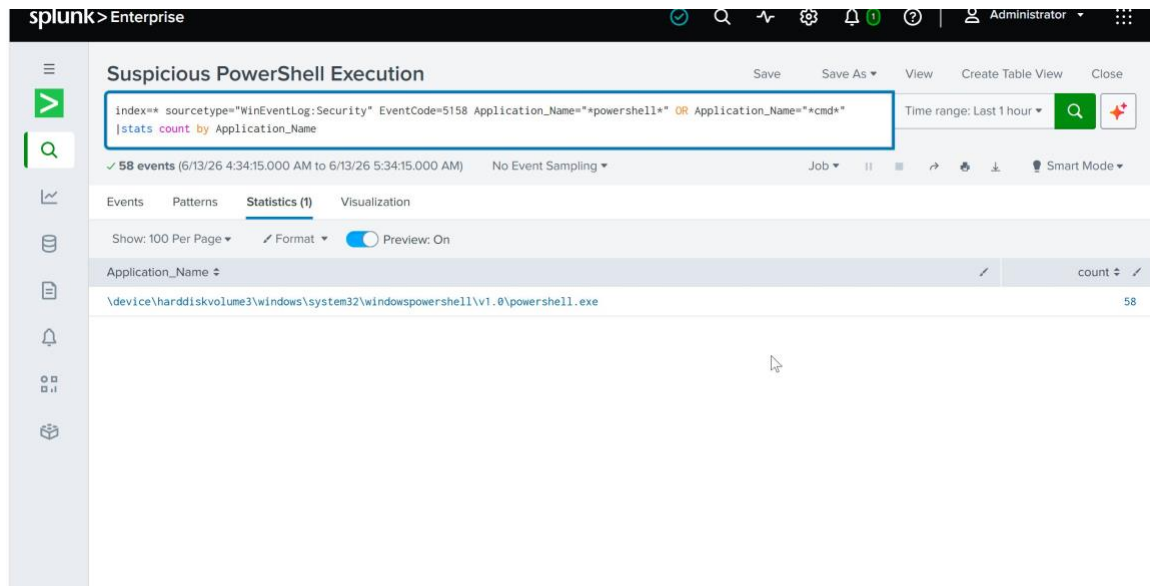


Figure 7: Connection-binding events attributed to powershell.exe - 58 events

6. Alert Configuration

Each detection query was converted into a scheduled Splunk alert. All three alerts are enabled and run on an hourly schedule, providing continuous monitoring coverage for the three simulated attack techniques.

Alert	Trigger Condition	Schedule	Severity	Status
Brute Force Attack - Failed Logins	Failed logons (4625) per account > 3	Hourly, at 15 min past hour	High	Enabled
Suspicious PowerShell Execution	PowerShell process creations (4688) > 0	Hourly	Medium	Enabled
Suspicious Port Scan - PowerShell	Connection binds (5158) from powershell/cmd > 0	Hourly	High	Enabled

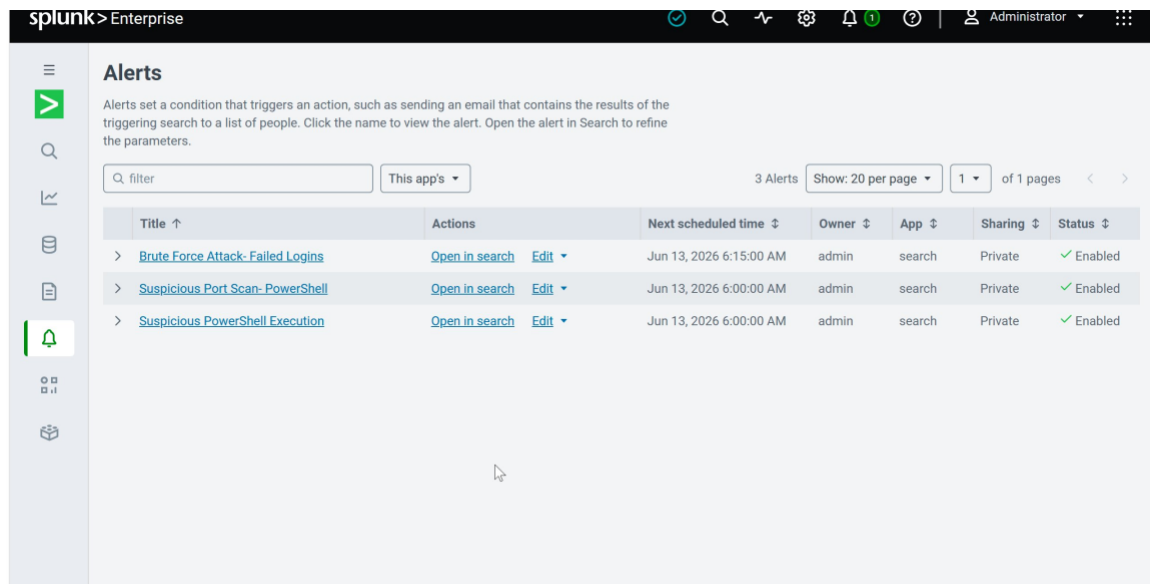


Figure 8: All three alerts configured and enabled in Splunk

7. SOC Monitoring Dashboard

A Classic Dashboard (“SOC Monitoring Dashboard”) was built to provide a single-pane view of all three detections. Each panel is driven directly by the same SPL query used in its corresponding alert, ensuring the dashboard and the alerting logic remain consistent.

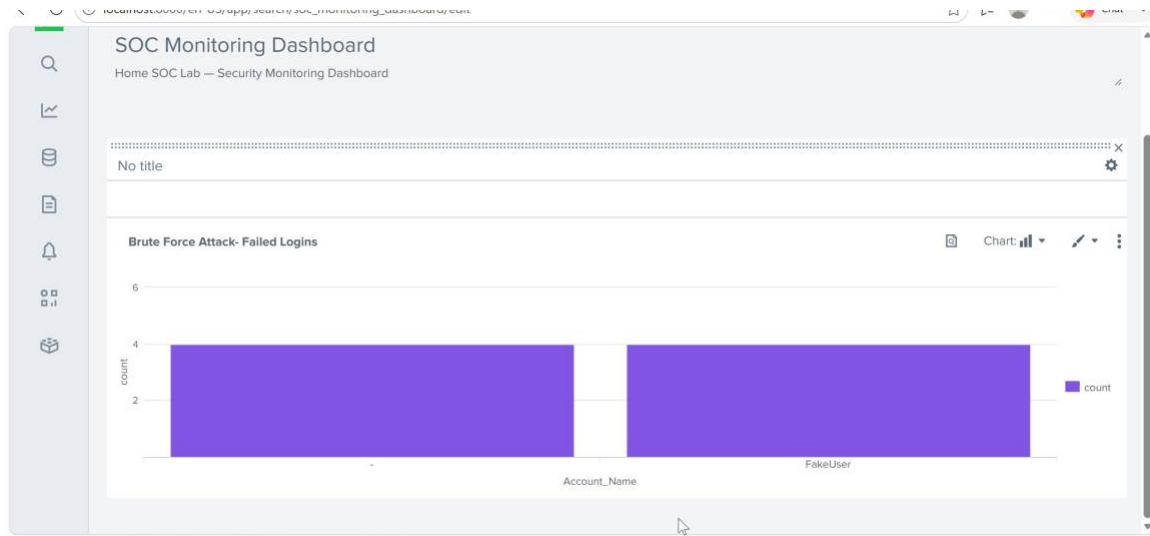


Figure 9: Dashboard panel - Brute Force Attack: Failed Logins by account (FakeUser: 4)

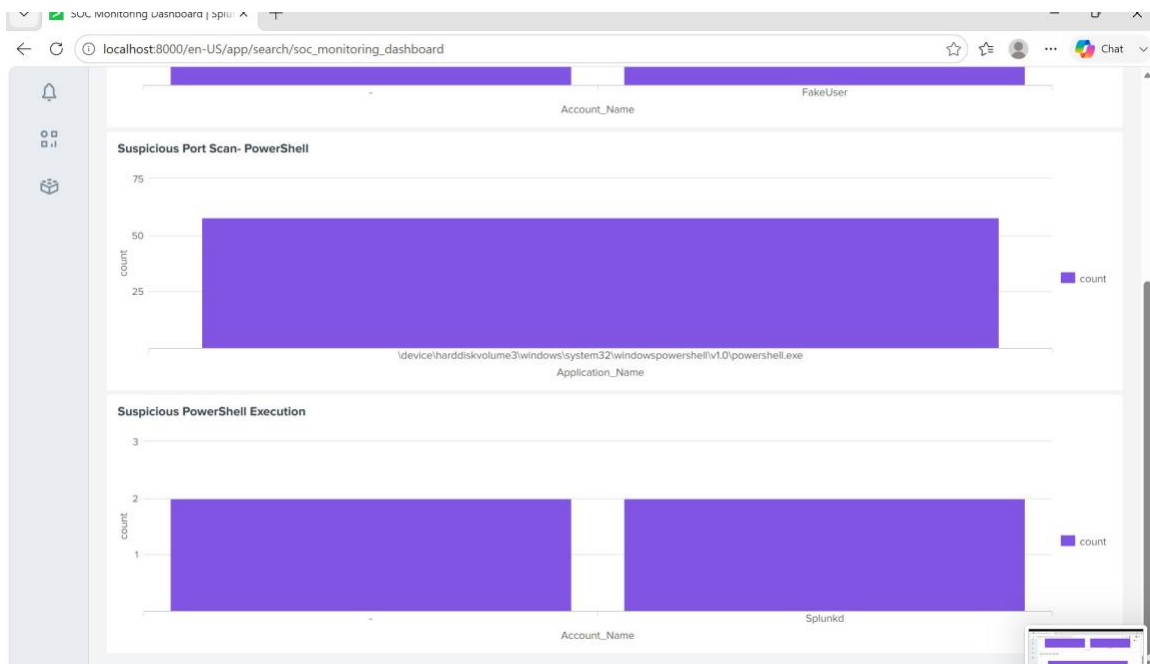


Figure 10: Dashboard panels - Suspicious Port Scan (powershell.exe: ~58 bindings) and Suspicious PowerShell Execution (taifs / Splunkd)

8. Incident Investigation & Timeline

By correlating the timestamps of the three detections, a chronological attack timeline was reconstructed:

Time (UTC)	Event Code	Source	Description
04:18:43	4625	FakeUser	Failed logon attempt #1 (brute force)
04:19:19	4625	FakeUser	Failed logon attempt #2
04:21:18	4625	FakeUser	Failed logon attempt #3
04:21:53	4625	FakeUser	Failed logon attempt #4 ,Alert 1 triggers (> 3 failures)
04:26:23	4688	taifs	PowerShell launched with -ExecutionPolicy Bypass; reconnaissance commands executed (whoami, hostname, ipconfig)
04:32:11	4688	taifs	Second PowerShell execution , Alert 2 triggers
04:44:46 - 04:47:44+	5158	powershell.exe	Sequential port-binding activity (~58 ports) consistent with port scan reconnaissance , Alert 3 triggers

8.1 Investigation Narrative

Analysis of the collected logs revealed a sequence of events consistent with a potential attack progression. The activity began with multiple failed authentication attempts targeting the account FakeUser, resulting in four failed logon events within a three-minute period. This pattern may indicate a brute-force or password-guessing attempt.

Shortly afterward, activity was observed under the account taifs, where PowerShell was executed with the execution policy bypassed. Commands such as whoami, hostname, and ipconfig were executed, which are commonly associated with host reconnaissance and post-compromise enumeration activities.

Approximately 12-18 minutes later, a significant increase in network-related events was detected. The host initiated port-binding activity across numerous ports, generating hundreds of network events. This behavior is consistent with service discovery or network reconnaissance techniques commonly used by attackers to identify accessible services and potential lateral movement opportunities.

By correlating authentication, process creation, and network activity logs, the investigation demonstrated how multiple security events can be linked to form a coherent attack timeline. The observed techniques align with several MITRE ATT&CK tactics, including Credential Access, Discovery, and Execution.

9. MITRE ATT&CK Mapping

Each detection scenario was mapped to its corresponding MITRE ATT&CK tactic and technique, framing the lab activity within an industry-standard threat model:

Detection Scenario	MITRE ATT&CK Tactic	Technique (ID)
Brute Force Authentication (Event ID 4625)	Credential Access	Brute Force (T1110)
Suspicious PowerShell Execution (Event ID 4688)	Execution / Discovery	Command and Scripting Interpreter: PowerShell (T1059.001); System Information / Network Configuration Discovery (T1082 / T1016)
Port Scan / Service Discovery (Event ID 5158)	Discovery	Network Service Discovery (T1046)

10. Conclusion & Recommendations

This project demonstrates an end-to-end SOC workflow: configuring log sources and auditing policies, simulating realistic attack techniques, developing and validating SPL-based detection logic, converting detections into scheduled alerts, consolidating findings into a monitoring dashboard, and reconstructing an incident timeline mapped to MITRE ATT&CK. Across the three scenarios, all detections correctly identified the simulated activity with no false positives observed in the test data.

10.1 Recommendations for Future Work

- Integrate Sysmon for richer process, network, and registry telemetry beyond native Windows auditing.
- Expand detection coverage to include lateral movement (e.g., PsExec, WMI) and credential dumping (e.g., LSASS access) scenarios.
- Add a fourth dashboard panel visualizing the end-to-end event timeline to support faster triage.
- Implement automated response actions (e.g., account lockout notification) for high-severity alerts.
- Extend log correlation to Elastic Stack (ELK) to practice cross-platform SIEM analysis.